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I am a postdoctoral researcher at ENS Lyon in the CASH team (Inria), under the supervision of Nicolas Tabareau and Cyril Cohen. My research focuses on AI-assisted formal reasoning and software verification in the Rocq and Lean proof assistants. I combine model–prover interaction, retrieval, autoformalisation, proof translation, tactic recommendation, and scalable infrastructure for training and evaluating formal-reasoning systems.

My current research includes:

- **Babel-formal**, which translates proofs between Lean and Rocq using proof terms as a pivot representation and also transfers proof styles, such as vanilla Rocq to MathComp’s SSReflect tactics. The work was presented at the MATH-AI workshop at Math-AI@NeurIPS 2025.
- **LLM4Docq**, which combines LLM-generated documentation with expert feedback to build natural-language documentation, retrieval, and autoformalisation resources for MathComp. The work was presented at [Rocqshop@ITP](https://rocqshop.itp.fr/) 2025
- **Tacq**, a context-aware tactic recommender that enriches Rocq based on dependency, notation, and natural-language information. Tacq was published in the JFLA 2026 proceedings.
- **Crrrocq**, an interactive theorem-proving system combining retrieval, reasoning, and Rocq feedback to construct proofs incrementally.
- **Pile-of-Rocq and rocq-ml-toolbox**, a data-and-infrastructure stack for machine learning with Rocq. Pile-of-Rocq is a structured corpus of sources, declarations, proofs, proof steps, dependencies, axioms, and documentation extracted from 48 diverse Rocq environments. It currently contains more than 47 million records. Its companion **rocq-ml-server** provides scalable parallel Rocq interaction for model inference, benchmark evaluation, trajectory generation, and reinforcement-learning or verifier-feedback workflows.

During my PhD (2019–2024), I spent one year in industry (2023–2024) as a Research Scientist in generative AI at Jumbo Mana where I designed and deployed AI pipelines for cultural and interactive applications, including speech recognition (ASR), speech synthesis (TTS), and large language models. This gave me experience in building AI systems that work in real-world contexts, with attention to robustness, usability, and user experience.

4.B Previous Experience

4.B.1 Professional Experience

Jan 2025 - Present | Postdoctoral Researcher

CASH Team, ENS Lyon (Inria) - Supervisors: Cyril Cohen, Nicolas Tabareau

- Research on AI-assisted formal reasoning in Rocq and Lean.
- Co-development of Crrrocq, Babel-formal, LLM4Docq, and Tacq, covering interactive proving, cross-assistant translation, documentation/retrieval, and tactic recommendation.
- Development of rocq-ml-toolbox and rocq-ml-server for scalable model–prover interaction, parallel inference, benchmark evaluation, trajectory generation, and reinforcement-learning workflows.
- Led a workgroup at Rocq’n’Share 2025 and co-organisation of Rocq’Band community meetings.
- Participation in the "Lean Wit It, Rocq Wit It" project funded by Renaissance Philanthropy.

Sep 2019 – Sep 2024 | Ph.D. in Mathematics

University Paris-Saclay, Laboratoire de Mathématiques d’Orsay - Supervisor: Christian Gérard

- Thesis: *Hadamard states for Dirac fields on curved spacetime*.
- Developed advanced mathematical tools in PDE, microlocal analysis, and quantum field theory.
- Published in **Letters in Mathematical Physics** and **Reviews in Mathematical Physics**.

Jan 2023 - Feb 2024 | Research Scientist in Generative AI

Jumbo Mana, Strasbourg

- Co-led “**Bonjour Vincent**” (Orsay Museum, Amazon Alexa), an AI-powered cultural mediation avatar.
 - Co-developed “**Felon-E**” (Gamescom, Paris Games Week), an interactive AI-based game.
 - Designed and deployed AI pipelines for multi-modal human-machine interaction in cultural contexts.
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4.B.2 Teaching Experience

Aug-Sep 2026 | Invited Lecturer Summer School “Proof Assistants and Applications” - University of Strasbourg

- Designed and delivered the course “Introduction to transformers and their application to proof assistants”.
- Delivered two 90-minute lectures and a two-hour hands-on tutorial for PhD students and researchers.
- Covered transformer architectures, large-language-model training, test-time reasoning, and applications to interactive theorem proving.

Sep 2019 – Sep 2022 | Teaching Assistant - Polytech Paris-Saclay

- 192 hours of undergraduate mathematics teaching (+100 students) with Associate Professor Rémi Leclercq
- Covered calculus, linear algebra, and applied mathematics for engineering students

Sep 2018 – Jul 2020 | Oral Examiner (“Colleur”) in Mathematics - Classe préparatoire MPSI/MP, Lycée Charlemagne, Paris

- Conducted weekly (2h/week) oral examinations for competitive entrance preparation to French Grandes Écoles
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4.C Education

4.C.1 Main Education

- **Ph.D. in Mathematics (defense: 21/03/2024)** - University Paris-Saclay, France (2019–2024)
Hadamard states for Dirac fields on curved spacetime - Supervisor: Christian Gérard.
- **ENS Paris-Saclay Diploma** (2019)
- **M2 AMS - Analysis, Modelling, Simulation** - University Paris-Saclay (2018–2019)
- **M2 MEEF - Agrégation in Mathematics** - ENS Paris-Saclay (2017–2018) - National competitive examination for teaching in higher education and preparatory classes (national rank: 24)

4.C.2 Additional Courses and Training

- NVIDIA Deep, *Model Parallelism: Building and Deploying Large Neural Networks* (2023)
 - *Type Theory* (M2MA, ENS Lyon), D. Pous
 - *Mathematical Components Lectures* (M2MA, ENS Lyon), C. Cohen
 - *Lean Proof Assistant Lectures* (Ph.D. level, ENS Lyon), F. Nuccio
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4.D.1 Awards and Grants

- **2024** - Special Prize for Technical Innovation, *Privacy-preserving AI Hackathon* (Zama x Entrepreneur First x Hugging Face, 50 participants)
- **2022** - Ph.D. Award, Fondation Mathématiques Jacques Hadamard (FMJH)
- **2016/2017** - Excellence Grant, Fondation Mathématiques Jacques Hadamard (FMJH)

4.D.2 Participation in Funded Projects

- **Postdoctoral Researcher**, *Lean Wit It, Rocq Wit It*, National project (France) led by N. Tabareau, funded by Renaissance Philanthropy (\$1M, 2025–2028)

4.E Scientific Output

4.E.1. Publications

1. J. Vienneot, G. Baudart, E. J. Gallego Arias, M. Lelarge, T. Stoskopf, *Tacq – Context Aware Tactic Recommendation for Rocq*, JFLA 2026, hal-0542814.
2. T. Stoskopf, C. Cohen, N. Tabareau, *Babel-formal: Translation of Proofs between Lean and Rocq*, MATH-AI@NeurIPS 2025, hal-05342510.
3. T. Stoskopf, J. Vienneot, C. Cohen, *LLM4Docq: Bootstrapping Documentation for MathComp with LLMs and Expert Feedback*, Rocqshop@ITP 2025, hal-05342426.
4. C. Gérard, T. Stoskopf, *Hadamard property of the in and out states for Dirac fields on asymptotically static spacetimes*, Lett. Math. Phys. (2022), DOI: 10.48550/arXiv.2108.11955, 10.1007/s11005-022-01556-9.
5. C. Gérard, T. Stoskopf, *Hadamard states for quantized Dirac fields on Lorentzian manifolds of bounded geometry*, Rev. Math. Phys. (2022), DOI: 10.48550/arXiv.2108.11630, 10.1142/S0129055X22500088.

4.E.4. Research software and datasets

6. J. Vienneot, G. Baudart, E. J. Gallego Arias, M. Lelarge, T. Stoskopf, *Goal2Tac*, github.com/LLM4Rocq/goal2tac (2026)
7. T. Stoskopf, *Pile-of-Rocq*, github.com/LLM4Rocq/pile-of-rocq (2026)
8. T. Stoskopf, *Rocq-ml-toolbox*, github.com/LLM4Rocq/rocq-ml-toolbox (2026)
9. T. Stoskopf, C. Cohen, N. Tabareau, *Babel-formal*, github.com/LLM4Rocq/babel-formal (2025)
10. T. Stoskopf, J. Vienneot, G. Baudart, M. Lelarge, *Crrrocq*, github.com/LLM4Rocq/crrrocq (2025)
11. T. Stoskopf, J. Vienneot, C. Cohen, *LLM4Docq*, github.com/LLM4Rocq/LLM4Docq (2025)
12. P. Galissant, T. Stoskopf, *ZLlamask*, github.com/theostos/ppai-hackathon (2024)

4.E.5. Software Output

- *Bonjour Vincent*, AI cultural avatar (Amazon Alexa 2024, Orsay Museum 2023)
- *Felon-E*, AI-based game (Gamescom 2023, Paris Games Week 2023)
- *Emmanuel Delran's Avatar* (CMA-CGM 2024)

4.E.6. Selected scientific presentations

- **2026** -IA and Mathematics Days: AI, Proof and Formalization Days, ENS Lyon, hands-on session. Designed and delivered a hands-on session on AI-assisted formal reasoning.
- **2025** - [Rocqshop@ITP](#), Reykjavik, Iceland, “LLM4Docq”.
- **2025** - Rocq’n’Share, Inria Paris, “Project-wide automatic documentation”. Led a workgroup on project-wide automatic documentation.
- **2025** - LCIS Scientific Day, Grenoble INP-UGA. Introduced the robustness guarantees provided by formal methods and discussed their potential impact on the reliability of generative models.
- **2025** - AI and Mathematics Conference, PSL University Paris, hands-on session. Designed and delivered a hands-on session introducing AI-assisted proof tools to a broader mathematics and AI audience.

4.F.1 Outreach and Public Engagement

- **2023–2024** - Co-led *Bonjour Vincent*. Co-designed and deployed an interactive generative-AI avatar that allowed visitors to converse with Vincent van Gogh. The installation was available throughout the exhibition *Van Gogh in Auvers-sur-Oise: The Final Months*, which attracted 794,000 visitors.
- **2024** - Piscine AI, POC Innovation. Introduced more than 50 engineering students to current AI research questions, methods, and experimental practices.
- **2025** - Tournée Génération IA (AI Summit Action), Mines Paris-PSL. Contributed to a MathAData hands-on activity for high-school students (+20), using accessible mathematical examples to explain how AI systems work and to discuss their applications and limitations.

4.G.1 Organizing committees

- **2025-Present** – Rocq’Band, ENS Lyon. Co-organizes twice-monthly meetings focused on disseminating tools from the Rocq ecosystem, presenting ongoing developments, and introducing Rocq to new users and research communities.
- **2020-2022** – Séminaire des doctorants LMO, Paris-Saclay University. Coordinated the twice-monthly seminar program, invited and scheduled external speakers, and secured financial support from the Société Mathématique de France (€2000).